

Computer Networks Spring 2026

Assignment#2 (6A & 6B)

Due Date: 19th February, 2026

Submission Mode & Time: Handwritten solutions to be submitted during the lecture.

Please note the following:

1. No exceptions to the above date and time will be allowed. Inability to submit the assignment by the required time will result in zero marks.
2. To ensure self-completion of assignments and discourage plagiarism, the instructor or the relevant TA may randomly contact you and ask for an explanation of your answers. Where plagiarism and/or cheating is evident, you will be referred to the departmental disciplinary committee. In extreme cases of plagiarism an F may be awarded immediately with further referral to the university disciplinary committee.
3. All solutions must be **hand-written**.
4. **Assignment Solution Submission:** In case of **in person / physical lectures at the campus**, hard copy of the hand-written assignment's solutions will be submitted by **hand** by each student to the Instructor / TA directly during the lecture on the due date.

Question: 01 [5+5+5+5 = 20 Marks] [CLO 2]

Suppose within your Web browser you click on a link to a Web page. The IP address for the associated URL is not cached in your local host, so DNS lookup is necessary to obtain the IP address. Suppose that 5 DNS servers are visited before your host receives the IP address from DNS. The successive visits incur an RTT (round-trip time) of 20 ms, 10 ms, 25 ms, 15 ms, and 30 ms respectively for the DNS servers. Suppose that the Web page associated with the link contains exactly one object, consists of a small amount of HTML text and incurs an RTT of 20 ms between the local host and the server containing the object.

- (a) Assuming zero transmission time of the object, how much time elapses from when the client clicks on the link until the client receives the object?
- (b) Refer to the same scenario as in Part (A), now suppose that the base HTML file (incurring RTT as 20 ms) references 5 small objects on the same server. The referenced objects incur the same amount of RTT as the base HTML file between the local host and the server containing the object. Neglecting transmission times, how much time elapses with:
 - i. **Non-persistent HTTP with no parallel TCP connections?**
 - ii. **Non-persistent HTTP with the browser configured for 3 parallel connections?**
 - iii. **Persistent HTTP with pipelining?**

Question: 02 [2+6+10 = 18 Marks] [CLO 2]

Suppose you start a new IT company called Fast Burgers and you decide to have an online presence. People typing either the URL fastburgers.com or fastburger.com into their web browser, both should land on your webserver. Your employees can be reached via email in the format, i.e. someone@fastburgers.com

For this, you install multiple machines (servers etc.) providing different services as listed below:

Sr#	Host Name (Server / Domain Name)	Type	IP Address
1	fastburgers.com	Webserver	20.21.22.23
2	smtp.fastburgers.com	Email Server	20.21.22.24
3	authoritative.fastburgers.com	Authoritative Name Server	20.21.22.25
4	database.fastburgers.com	Backend Database (Containing customer and employee information)	20.21.22.26

In order to get your online network operational and accessible on the Internet, please answer the following questions.

(Note: Please use the following format for resource record (RR) and ignore the TTL field.)
RR Format: (name, value, type)

- a) List the DNS resource records (RR) to be added to the root name servers.
- b) List the DNS resource records (RR) to be added to the TLD name servers. Who does that?
- c) List all the DNS resource records (RRs) to be added to your authoritative name server.

Question: 03 [1 x 5 = 5 Marks] [CLO 2]

Below is a server to client HTTP Response Message.

HTTP/1.0 404 Not Found
Date: Tue, 07 May 2024 11:33:45 +0000
Server: Apache/2.2.3 (CentOS)
Content-Length: 258
Connection: Close
Content-type: text/html

Please answer the questions given below:

- i) Was the server able to send the file (object) successfully?
- ii) Is the HTTP connection persistent or non-persistent?

- iii) What is the type of the file (object) being sent?
- iv) What is the size of the file (object) being sent in bytes?
- v) What is the name & version of the server?

Question: 04 [2+3+2 = 7 Marks] [CLO 2]

An institutional network is connected to the Internet via an access link with a bandwidth of 15 Mbps.

- The average object size is 100 Kbits.
 - The average request rate from the institution's browsers to origin servers is 100 requests/second.
 - The RTT from the institutional router to the origin server is 2 seconds.
 - The LAN delay is negligible.
- a) Calculate the traffic intensity (utilization) on the access link. What does this imply for delay?
- b) The institution installs a Web Cache (Proxy Server). Suppose the cache hit rate is 0.4 (40%). Calculate the new arrival rate of requests to the access link and the resulting link utilization.
- c) Assuming the cache lookup takes 10 ms (0.01s) and the access link delay becomes negligible when utilization is below 0.8, what is the new average end-to-end delay?